

CHRISTOPHER WANG

INFORMATION

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EDUCATION

Cornell University	Ithaca, NY
M.S. Mathematics	May 2025
Ph.D. Mathematics, in progress	August 2022–present
Advisor: Alex Townsend	

Columbia University	New York, NY
B.A. Mathematics, summa cum laude with departmental honors	May 2022
Advisor: Ioannis Karatzas	
Thesis: <i>Independence and conditional independence in discrete-time dynamic allocation problems</i>	

RESEARCH VISITS

Argonne National Laboratory	Lemont, IL
Givens Associate under Wendy Di	June–August 2026

Institute for Pure and Applied Mathematics (IPAM)	Los Angeles, CA
Workshop on Randomized Numerical Linear Algebra	August 2025
“Randomized algorithms for solving inverse problems in X-ray science” under Wendy Di and Sherry Li	

Institute for Computational and Experimental Research in Mathematics (ICERM)	Providence, RI
Semester program, “Numerical PDEs: Analysis, Algorithms, and Data Challenges”	January–May 2024

PUBLICATIONS & PREPRINTS

- Query complexity lower bounds for operator learning with advective PDEs (with Diana Halikias and Alex Townsend), in preparation.
- Oblivious subspace injection is not enough for relative error (with Alex Townsend). [\[arXiv:2604.10215\]](#)
- Beyond singular value gaps in randomized subspace approximation (with Alex Townsend). [\[arXiv:2603.01191\]](#)
- Convergence of pivoted Cholesky for Lipschitz kernels (with Sungwoo Jeong and Alex Townsend). [\[arXiv:2509.13582\]](#)
- The Gittins index is optimal for dynamic allocation with conditionally independent filtrations. *Advances in Applied Probability*, to appear. [\[arXiv:2312.09350\]](#)
- Operator learning for hyperbolic partial differential equations (with Alex Townsend). *Journal of Machine Learning Research* 26(199), 1–44, 2025. [\[arXiv:2312.17489\]](#)
- Extensions of true skewness for unimodal distributions (with Yevgeniy Kovchegov, Alex Negrón, and Clarice Pertel). *Mathematical Methods of Statistics* 33(3), 239–258, 2024. [\[arXiv:2209.11139\]](#)

GRANTS & AWARDS

- NSF Graduate Research Fellowship 2023–2028
- RTG Graduate Fellowship, Cornell University 2022–2023
- John Dash van Buren, Jr. Prize in Mathematics, Columbia University 2022
Awarded to 1–2 graduating students for outstanding senior theses
- John Jay Scholar, Columbia University 2018–2022

SELECTED TALKS & PRESENTATIONS

- *Query complexity lower bounds for operator learning with advective PDEs* (invited talk) February 2027
SIAM Conference on Computational Science and Engineering (CSE) Pittsburgh, PA
- *Beyond the singular value gap for randomized subspace approximation* (invited talk) November 2026
SIAM Conference on Mathematics of Data Science (MDS) Salt Lake City, UT
- *Unknown hierarchies, hyperbolic PDE, and randomized rank detection* (invited talk) November 2026
AMS Fall Western Sectional Meeting Tempe, AZ
- *Beyond the singular value gap for randomized subspace approximation* October 2026
SIAM NY–NJ–PA Section Conference Piscataway, NJ
- *Query complexity lower bounds for operator learning with advective PDEs* (invited talk) October 2026
Math Frontier Seminar, Rensselaer Polytechnic Institute Troy, NY
- *Unknown hierarchies, hyperbolic PDE, and randomized rank detection* (invited talk) May 2026
27th Conference of the International Linear Algebra Society (ILAS) Blacksburg, VA
- *Operator learning for hyperbolic PDEs* (invited talk) May 2026
Mathematical AI Seminar, Fields Institute (online)
- *Advantages and disadvantages of technology in the math classroom* December 2025
Department of Mathematics Teaching Seminar, Cornell University Ithaca, NY
- *Beyond the singular value gap for randomized subspace approximation* November 2025
Scientific Computing & Numerics (SCAN) Seminar, Cornell University Ithaca, NY
- *Operator learning for hyperbolic PDEs* (invited talk) November 2025
SIAM NY–NJ–PA Section Conference University Park, PA
- *Exact distribution functions for randomized subspace iteration* (poster) June 2025
Householder Symposium XXII Ithaca, NY
- *Operator learning for hyperbolic PDEs* (poster) October 2024
SIAM Conference on Mathematics of Data Science (MDS) Atlanta, GA
- *PDE learning beyond neural nets* (invited talk) April 2024
Applied Mathematics (APMA) Graduate Student Seminar, Brown University Providence, RI
- *Operator learning for hyperbolic PDEs* (poster) February 2024
ICERM Workshop: Numerical Analysis of Multiphysics Problems Providence, RI
- *Extensions of true skewness for unimodal distributions* (poster) April 2022
Joint Mathematics Meeting, Pi Mu Epsilon Student Poster Session (online)

TEACHING

Cornell University

- Teaching Assistant, MATH 2310: Linear Algebra for Data Science Fall 2025–Spring 2026

Columbia University

- Undergraduate Teaching Assistant, MATH 4061: Intro to Modern Analysis I Spring 2022
- Undergraduate Teaching Assistant, MATH 1011: Calculus I Fall 2021

SERVICE

Referee for SIURO

2026

Outreach Co-chair for Cornell Association for Women in Mathematics

Fall 2022–Spring 2026

Planned several AWM outreach events, notably the 2023–2026 Cornell Julia Robinson Math Festivals for K–12 students, featuring various volunteer-led math activities and a speaker event aimed toward children and general audiences. Also organized the math workshop at Cornell’s 2025 Expanding Your Horizons event for middle school girls, as well as the 2024–2025 AWM ZigZag mentoring program.

Directed Reading Program Mentor

Fall 2022–Spring 2023, Fall 2024

Mentored three undergraduate students at Cornell in directed reading in differential topology, partial differential equations, and probability theory.

Volunteer for Cornell Little Math Circle

Fall 2023

Led several sessions of math activities for kindergarten through 5th grade students.

Volunteer for Cornell Big Red Math Competition

Fall 2023

Proctored and graded exams for a high school math competition hosted by Cornell.

Graduate Assistant for the Cornell Math REU

Summer 2023

Advised four undergraduate students participating in the REU program at Cornell in the area of numerical PDEs. Activities included lectures on background theory and weekly meetings with project teams.

Graduate Mentoring Program Co-organizer

Spring 2023–Fall 2023

Matched matriculating students with peer mentors and organized mentor-mentee social events.

MISCELLANEOUS

- Spoken languages: English (native), Mandarin Chinese (native), French (elementary), Spanish (elementary)
- Programming languages: MATLAB, Python, Javascript, Java, C

REFERENCES

- Alex Townsend [townsend@cornell.edu]
Associate Professor
Cornell University, Department of Mathematics
- Zichao Wendy Di [wendydi@mcs.anl.gov]
Computational Mathematician
Argonne National Laboratory, Mathematics and Computer Science Division & X-ray Science Division
- Ioannis Karatzas [ik@math.columbia.edu]
Eugene Higgins Professor of Applied Probability
Columbia University, Department of Mathematics
- Yunan Yang (teaching) [yunan.yang@cornell.edu]
Goenka Family Assistant Professor
Cornell University, Department of Mathematics
- Marie MacDonald (teaching) [mb2636@cornell.edu]
Associate Teaching Professor & Stephen H. Weiss Provost Teaching Fellow
Cornell University, Department of Mathematics